

Prosumer Oriented Energy Sharing Model for Grid Integrated Electric Rickshaw Charging Garages

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Abstract- The growing popularity of electric rickshaws in cities requires implementing cost and environment efficient charging systems. The need to develop integrated renewable energy systems can be supported by intense peak loads and variable energy prices that typify traditional, grid only charging stations. In this paper, HOMER Pro is used to simulate a prosumer oriented energy sharing system of electric rickshaw charging garages. The suggested system creates a hybrid energy system, which integrates solar photovoltaic (PV) arrays with the domestic electrical grid. Under such a set up excess electricity is exported back into the grid, with the PV installation being used as the main source of auxiliary charging loads. The generation of solar energy is supplemented by energy imported to the grid at times of inadequate solar generation so that there is no interruption. According to simulation, this strategy helps to reduce greenhouse gas emissions and overall operational costs, and, at the same time, it helps to eliminate reliance on the traditional grid. Additionally, the two way power flow will allow the charging station to operate as a participator in the energy market as an active prosumer, and this power will improve energy reliability. The promoted framework indicates that renewable integration improves the creation of the smart grid ecosystems and suggests the solution to sustainable urban transportation on a scale.

Keywords: Prosumer, Energy sharing, Electric rickshaw charging, Grid integration, Renewable energy.

1. INTRODUCTION

The need for effective and environmentally friendly charging infrastructure has grown because of the electric rickshaws' explosive rise in popularity in cities. Issues with traditional grid dependent garages include high peak loads, erratic supply, and high operating costs. Solar photovoltaic (PV) panels provide the primary energy for charging stations and auxiliary loads in this paper's prosumer oriented energy sharing model for electric rickshaw charging garages, which is simulated using HOMER Pro. Reliability, cost savings, and carbon emissions can be achieved by exporting excess electricity to the grid and importing more energy as needed. The framework encourages smart city development and sustainable urban mobility. To increase parking efficiency and space utilization, prior research on smart parking has concentrated on sensor based detection, real time monitoring, and digital platforms [1]. Lan et al. [2] claim that previous studies highlight the difficulties of incorporating variable renewable energy sources into regional power grids, particularly concerning variability, intermittency, and grid stability, and demonstrate that variable speed pumped hydro storage can mitigate these issues. The potential of intelligent, prosumer focused urban energy systems is demonstrated by strategies like variable-speed storage, intelligent charging control, and multi objective optimization, which increase flexibility, lower costs, and

improve energy management, as discussed in [3][4]. According to [5], simulations in the PowerMatchingCity smart grid investigate agent based and constrained optimization for EV charging, balancing real time pricing, battery health, and user mobility. Using PV based systems and Arduino control, this study assesses Plug in Hybrid Electric Vehicle(PHEV)/ Plug in Electric Vehicle(PEV) charging scenarios, emphasizing the effects on emissions, expenses, voltage profiles, and upcoming smart charging technologies [6]. Research on EV charging infrastructure prioritizes cost effectiveness, equitable scheduling, and smart scheduling while taking user needs, grid impact, and battery degradation into account. Basic AC outlets can accommodate most daily charging needs, while solar PV integration lowers emissions and energy expenses [8]. Reliability, user satisfaction, and sustainable urban mobility are all improved by optimized algorithms and infrastructure strategies [7] [9]. The Nice Grid project, which is a component of GRID4EU(Grid for European Union), shows how to integrate photovoltaic powered neighborhoods into the smart grid. It also evaluates how distributed energy resources affect the stability, dependability, and flexibility of the low voltage grid and tackles issues with renewable energy management and electricity supply[10]. From [11], research emphasizes time-of use pricing, multi charging systems, and adaptive EV charging techniques as ways to optimize operator utility and reduce grid impact. In dynamic, dispersed smart grid environments,

emerging techniques like machine learning and intelligent control improve cybersecurity, resilience, and efficiency, according to [12]. Recent studies have assessed the feasibility of hybrid solar photovoltaic and biogas generator based charging stations for electric rickshaws in developing countries, demonstrating the potential for renewable energy integration in urban mobility [13].



Fig. 1. Visualization of a solar equipped rickshaw garage.

Unregulated widespread EV charging raises expenses and jeopardizes grid stability. Urban energy systems now face both new opportunities and challenges because of the growing use of renewable energy sources and electric vehicles (EVs). One way to lessen the environmental impact of transportation, which contributes significantly to greenhouse gas emissions, is to electrify vehicles. However, the quick rise in EV use puts a lot of strain on the infrastructure needed for charging, necessitating clever, dependable, and affordable solutions. System stability, load management, and operational efficiency are further complicated when distributed energy resources like wind and solar photovoltaic (PV) panels are integrated into the power grid. Adaptive charging, demand response mechanisms, and prosumer oriented energy sharing are made possible by smart grid technologies, which include communication, control, and optimization strategies. Decisions about energy management, grid integration, and renewable energy use are supported by the detailed modeling of hybrid energy systems made possible by simulation tools such as HOMER Pro. With an emphasis on grid interaction, solar PV integration, and efficient energy management techniques, this paper proposes a prosumer oriented energy sharing model for electric rickshaw charging garages.

2.THE DESIGN

The project's prototype was created as a cohesive system that integrates algorithmic control and hardware operation. A battery storage unit for stability and backup, a bidirectional grid connection, solar photovoltaic panels as the main renewable source, and the electric rickshaw charging terminals as the main load are all part of the setup. To ensure dependable operation, power electronic converters and controllers are used to control and manage the energy distribution among these parts.

The algorithm flowchart, located on the left hand side of Fig. 2, outlines the system's sequential control sequence. All operational parameters, such as PV generation, battery state of charge (SOC), grid status, and charging demand, are set at the start of the process. After that, the system detects energy data in real time and assesses the supply of renewable energy. The electric rickshaws are charged using local energy sources if the PV and battery can handle the garage load. To guarantee energy stability, the battery's state of charge is updated. The system automatically switches to using more energy from the grid if the renewable supply is not enough. At every step, this iterative process is conducted to ensure dependability and continuous charging.

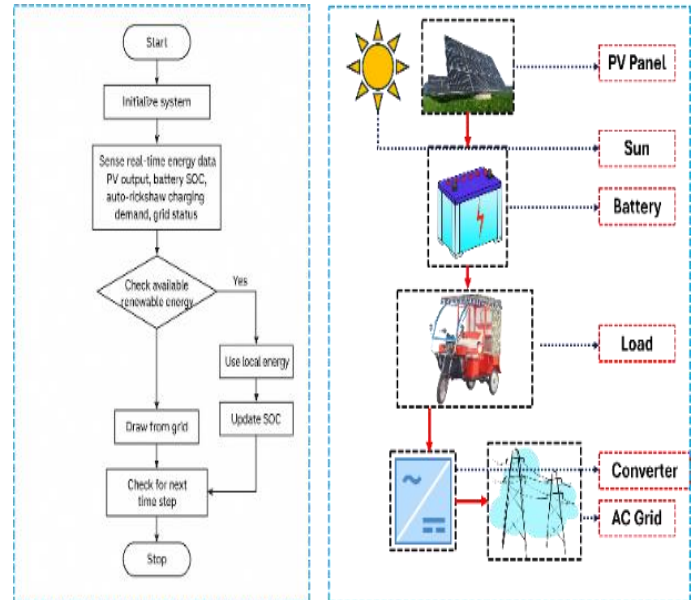


Fig. 2. Conceptual flow chart and block diagram of the system

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The hardware energy flow, which illustrates the physical integration of various components, is displayed on the right hand side of Fig. 2. Renewable energy produced by solar panels is either stored in the battery or sent straight to the charging terminals. When solar generation is low, the stored energy serves as a backup, and any shortfall is made up by importing electricity from the grid. The electric rickshaw garage serves as the primary load, and power electronic converters guarantee controlled energy flow between the subsystems.

2.1 Prototype Functional Workflow

A structured flowchart that clearly illustrates the operational logic of the prototype serves as a representation of the functional workflow of the suggested system. In order to maintain an effective balance between local renewable resources, storage, and the grid, this workflow demonstrates how various inputs, decision making procedures, and energy allocation strategies interact.

The energy sharing model's prototype functional workflow is shown in Fig. 3. If the demand exceeds the area's supply, the system requests support from the grid to ensure ongoing charging. Also, the system supports prosumer oriented energy sharing by exporting excess renewable energy to the grid whenever it becomes available. To guarantee ongoing monitoring and ideal energy management, the procedure concludes by updating the system status and loops for the following time step.

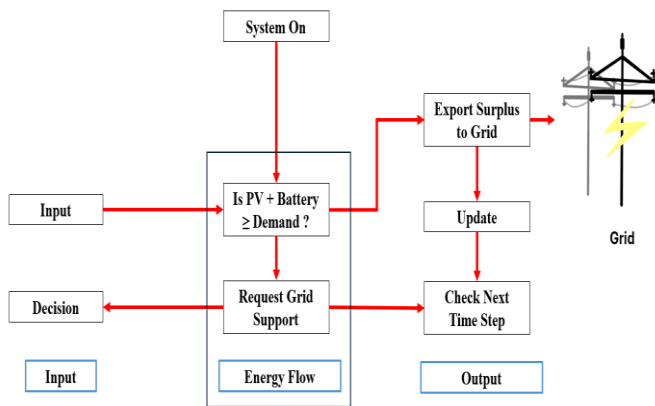


Fig. 3. Energy sharing model flowchart

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2.2 Software Implementation & Results

To evaluate the performance of the proposed solar PV system, a detailed simulation was carried out using HOMER Pro software. HOMER Pro is widely recognized for its ability to model hybrid renewable energy systems by optimizing component sizing, load management, and energy interactions with the grid. In this study, the software was used to simulate the energy production from the solar panels, battery storage performance, and grid interactions (both energy purchase and energy sale). The obtained results provide insights into system efficiency, grid dependency, and the economic feasibility of the design, which are later compared with the hardware implementation results for validation.

TABLE I. DAILY LOAD ESTIMATION

Components	Rating and Usage Pattern	Quantity	Energy Per unit (kWh/day)	Total Energy (kWh/day)
Rickshaw Charging	4.06 kWh/day each	10	4.06	40.60
LED Lights	10 W × 6 h	10	0.06	0.60
Mobile Charging	10 W × 2 h	10	0.02	0.20
Total	—	—	—	41.40

2.3 Load Estimation

The designed system considers three primary end uses: charging of ten electric rickshaws, lighting for ten units, and charging of ten mobile devices. Table I summarizes the daily energy requirements. So, the scaled average daily load is 41.4 kWh/day, resulting in an annual energy consumption of 15,111 kWh/year. The size of the renewable energy system and storage components are estimated on this basis. The system should be designed in such a way so that it can fulfil the loads, also the seasonal changes and efficiency losses are considered.

2.4 Hourly Load Profile

The load is distributed over a 24-hour cycle. Electric rickshaw charging is scheduled overnight (22:00–04:00), while lighting and mobile charging occur in the evening. The peak instantaneous demand is 6.87 kW during overlapping electric rickshaw charging and lighting hours. The hourly load distribution of the garage is shown in Table 3. Electric rickshaw charging occurs overnight (22:00–04:00), while lighting and mobile charging are concentrated in the evening hours. The peak demand of 6.87 kW occurs during overlapping electric rickshaw charging and lighting periods.

TABLE II. HOURLY LOAD DISTRIBUTION

Hour Range	Rickshaw (kW)	Lights (kW)	Mobiles (kW)	Total (kW)
0–3	6.77	0.10	0.00	6.87
4–6	6.77	0.00	0.00	6.77
7–9	0.00	0.00	0.00	0.00
10–13	2.50	0.00	0.00	2.50
14–15	0.00	0.00	0.00	0.00
16–17	0.00	0.18	0.00	0.18
18–21	0.00	0.43	0.00	0.43
22–23	6.77	0.10	0.00	6.87

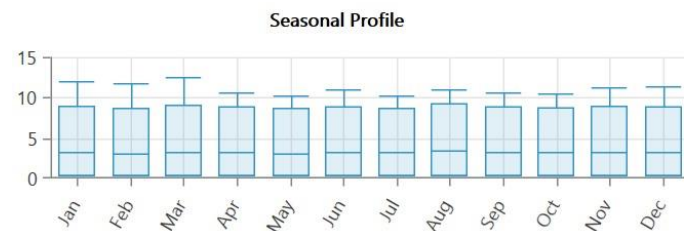


Fig. 4. Seasonal load distribution.

Here the seasonal profile in figure 4 illustrates the average daily energy demand which remains relatively stable throughout the year, where typical values are around 9 to 10 kWh/day and the highest peak reaches about 12 to 14 kWh/day.

This type of consistency simplifies the design and ensures reliable operation across all months.

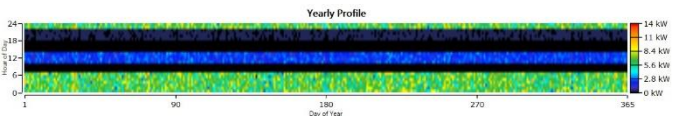


Fig. 5. Yearly load distribution.

In the figure 5, the yearly profile shows that the pattern of demand is very repetitive and stationary. Most of the demand is seen to be in the 2 to 6 KW range as the green regions are the dominant one. This figure proves that despite having some

increasing demand , the system load balance is predictable, which is beneficial in planning a renewable energy system.

3.ANALYSIS

Fig.6 presents a comprehensive comparison of four energy system cases, combining economic and operational metrics in a single figure. In case 1 a PV system was formed with a battery energy storage system to meet site demand and export surplus energy whereas case 2 is a grid only design meaning 100% reliance is on the utility grid. Case 3 is a design to maximize revenue, so the initial investment is high. Case 4 is fully dependent on the grid for primary power and on a diesel generator for backup power representing a high operating cost nonrenewable scenario.

TABLE III. GRID ENERGY PURCHASE AND SALE WITH NET EXCHANGE ANALYSIS

Case	NPC (₹)	LCOE (₹/kWh)	CAPEX (₹)	Operating Cost (₹/yr)	Grid Purchased (kWh)	Grid Sold (kWh)
1	1,303,920	0.739	8,825,203	-581,804	11,996	106,289
2	3,118,738	8.000	0	241,248	30,156	0
3	1.05E+07	5.949	1.43E+07	-295,147	11,996	106,550
4	1.24E+07	31.74	5,517,265	530,313	30,155	0

Table III compares four energy system cases in terms of economic and operational performance, including Net Present Cost (NPC), Levelized Cost of Energy (LCOE), capital expenditure (CAPEX), annual operating cost, and grid energy exchange. Case 1 shows the lowest LCOE (0.739 ₹/kWh) with high CAPEX but revenue from energy sold to the grid, while Case 2 requires no capital investment but depends entirely on grid purchases, resulting in a higher LCOE (8.000 ₹/kWh). Case 3 has the highest NPC and CAPEX yet benefits from selling energy back to the grid. Case 4 exhibits the highest LCOE (31.74 ₹/kWh) and operating cost, relying fully on grid purchases. This comparison highlights tradeoffs between investment, operational costs, and energy exchange for selecting an optimal system.

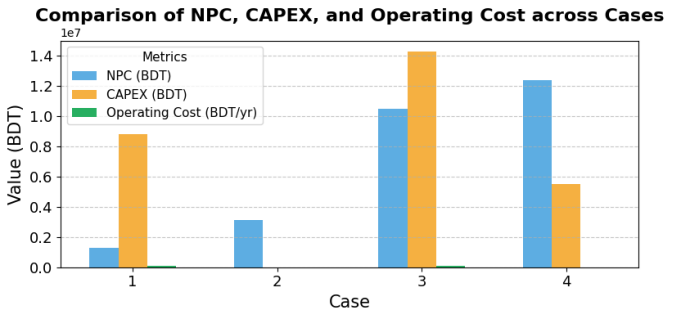


Fig. 6. Comparative representation of grid energy dynamics.

CAPEX (initial investment) is highest in Case 3, while Case 2 requires no capital investment. OPEX reflects operational costs and revenue: negative values (Cases 1 and 3) indicate net revenue from selling electricity to the grid. The bars for Grid Purchased and Grid Sold show how much energy each system buys from or sells to the grid, highlighting dependency and surplus generation. The line plot on the secondary axis represents LCOE, showing the cost of energy per kWh for each case. Case 1 achieves the lowest LCOE (0.739 ₹/kWh),

demonstrating cost efficiency despite high CAPEX, while Case 4 has the highest LCOE (31.74 ₹/kWh), reflecting high operating costs and full dependence on grid energy. Overall, the graph visually summarizes tradeoffs between investment, operational cost, energy exchange, and cost efficiency, making it easy to identify the most economically and operationally favorable system.

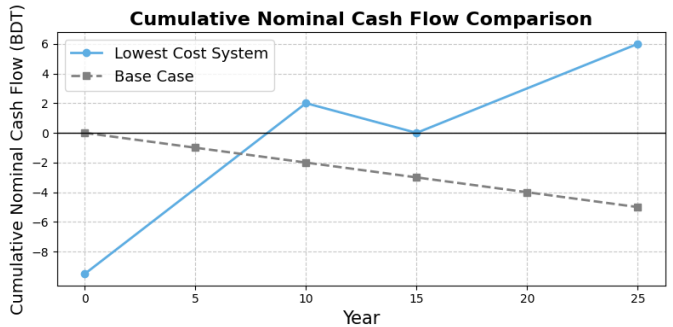


Fig. 7. HOMER representation of cumulative nominal cash flow.

Figure 7 illustrates how the hybrid system achieves positive cumulative cash flow over 25 years, outperforming the base case by reducing costs and providing long term economic savings.

4.DISCUSSION AND CONCLUSION

A hybrid PV battery grid system intended to effectively control energy flow is demonstrated in this project. The PV panel supplies connected loads and charges the battery at the same time. If more energy is required, it can be taken from the grid or sold to it. Simulation results highlight differences in generation and load, with Cases 1 and 3 achieving full renewable supply and Cases 2 and 4 depending partially on the grid and battery. By computing net energy exchange and grid dependency, software simulations enhance hardware and offer

information about system self sufficiency. Overall, the results validate the system's ability to optimize renewable usage, maintain load supply, and balance grid interaction, demonstrating both practical and analytical performance.

This was mainly a simulation based study but, in the future, proper hardware should be created and tested in real life situations for various conditions to confirm performance for different conditions. This project is an example of a hybrid system of energy management, which integrates interaction with the grid, battery storage and photovoltaic panels. This study shows a hybrid PV battery grid energy management system for electric rickshaw charging garages. The simulations successfully demonstrate the integration of renewable generation, energy storage, and grid interaction to create a prosumer-oriented energy sharing model. The major aim was to optimize the use of renewable energy sources and eliminate reliance on the grid and at the same time maintain continuity to the load connected to the renewable energy source. At the analytical stage, the HOMER Pro modelling system was adopted in which the photovoltaic (PV) generation, battery storage, and load demand of the rickshaw charging, light, and mobile charging were evaluated. The results of simulation highlighted the ability to balance renewable generation and storage of the system to ensure constant supply and measure the contribution and reliability of renewable energy. The imported energy to satisfy demand is reflected in negative net exchange values where excess energy sold back to the grid is reflected in positive values. This dynamically balanced generation, storage, and load of the system is supported by the built in analysis of hardware and software results. As a result, the project evidence that a hybrid PV battery grid system can competently handle energy, secure loads, maximize the use of renewable resources, and reduce grid dependency.

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7. NOMENCLATURE

Symbol	Meaning	Unit
PV	Photovoltaic	
NPC	Net Present Cost	
LCOE	Levelized Cost of Energy	
CAPEX	Capital Expenditure	
OPEX	Operational Expenditure	
PHEV	Plugin Hybrid Electric Vehicle	
PEV	Plugin Electric Vehicle	
EV	Electric Vehicle	
HOMER Pro	Hybrid Optimization of Multiple Energy Resources	